WORKED SOLUTION

STEP II 2025, Question 4

Pure •

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Everything rests on one fact: $\lfloor x \rfloor = m$ means exactly that m is an integer with $m \leq x < m+1$. Part (i) says that integer shifts pass straight through the floor. With that, f_n turns out to be periodic with period $\frac{1}{n}$, so we only need it on one short interval, where every term is easy to work out. Part (iii) is the case $n = 2$ in disguise, and it makes the infinite sum telescope.

Solution

Part (i)

Let $m = \lfloor x \rfloor$, so m is an integer and $m \leq x < m+1$. Adding the integer n gives

$$m + n \leq x + n < (m+n) + 1.$$

Since $m+n$ is an integer, it is the largest integer not exceeding $x+n$. So $\lfloor x+n \rfloor = m+n = \lfloor x \rfloor + n$, as required.

Part (ii)(a)

Write $\mathrm{f}_n(x) = \sum_{k=0}^{n-1} \lfloor x + \frac{k}{n} \rfloor - \lfloor nx \rfloor$. Replacing x by $x + \frac{1}{n}$ moves each term in the sum along by one place:

$$\mathrm{f}_n\left(x + \tfrac{1}{n}\right) = \sum_{k=1}^{n} \left\lfloor x + \tfrac{k}{n} \right\rfloor - \lfloor nx + 1 \rfloor.$$

The last term of the sum is $\lfloor x+1 \rfloor = \lfloor x \rfloor + 1$, and $\lfloor nx+1 \rfloor = \lfloor nx \rfloor + 1$, both by part (i) with $n = 1$. So

$$\mathrm{f}_n\left(x + \tfrac{1}{n}\right) = \lfloor x \rfloor + 1 + \sum_{k=1}^{n-1} \left\lfloor x + \tfrac{k}{n} \right\rfloor - \lfloor nx \rfloor - 1 = \mathrm{f}_n(x).$$

Part (ii)(b)

Let $0 \le t < \frac{1}{n}$. For $0 \le k \le n-1$,

$$0 \le t + \tfrac{k}{n} < \tfrac{1}{n} + \tfrac{n-1}{n} = 1,$$

so $\lfloor t + \frac{k}{n} \rfloor = 0$. Also $0 \le nt < 1$, so $\lfloor nt \rfloor = 0$. Every term is zero, and therefore $\mathrm{f}_n(t) = 0$.

Part (ii)(c)

> **Ask yourself** Part (a) only lets us step *up* by $\frac{1}{n}$. How do we reach an x that lies below $[0, \frac{1}{n})$?

Applying (a) at the point $x - \frac{1}{n}$ gives $\mathrm{f}_n(x) = \mathrm{f}_n(x - \frac{1}{n})$, so we can step down as well as up. By induction, $\mathrm{f}_n(x + \frac{m}{n}) = \mathrm{f}_n(x)$ for every integer m, positive or negative.

Now take any real x and let $m = \lfloor nx \rfloor$, so $m \le nx < m + 1$. Put $t = x - \frac{m}{n}$. Dividing by $n > 0$ gives $0 \le t < \frac{1}{n}$. Hence

$$\mathrm{f}_n(x) = \mathrm{f}_n\big(t + \tfrac{m}{n}\big) = \mathrm{f}_n(t) = 0$$

by (b). So $\mathrm{f}_n(x) = 0$ for all x, which is the identity

$$\lfloor x \rfloor + \left\lfloor x + \tfrac{1}{n} \right\rfloor + \cdots + \left\lfloor x + \tfrac{n-1}{n} \right\rfloor = \lfloor nx \rfloor.$$

Part (iii)(a)

Take $n = 2$ in (ii)(c) and evaluate at $\frac{x}{2}$:

$$0 = \mathrm{f}_2\big(\tfrac{x}{2}\big) = \left\lfloor \tfrac{x}{2} \right\rfloor + \left\lfloor \tfrac{x}{2} + \tfrac{1}{2} \right\rfloor - \left\lfloor 2 \cdot \tfrac{x}{2} \right\rfloor,$$

which rearranges to $\lfloor \frac{x}{2} \rfloor + \lfloor \frac{x+1}{2} \rfloor = \lfloor x \rfloor$.

Part (iii)(b)

> **Ask yourself** How can (iii)(a), which has a lone $+1$ in it, help with the term $\lfloor \frac{x+2^k}{2^{k+1}} \rfloor$?

Divide the top and bottom by 2^k. With $y = x/2^k$,

$$\left\lfloor \frac{x + 2^k}{2^{k+1}} \right\rfloor = \left\lfloor \frac{y+1}{2} \right\rfloor = \lfloor y \rfloor - \left\lfloor \frac{y}{2} \right\rfloor = \left\lfloor \frac{x}{2^k} \right\rfloor - \left\lfloor \frac{x}{2^{k+1}} \right\rfloor,$$

using (iii)(a) with y in place of x. So the sum of the terms for $k = 0, 1, \ldots, K$ telescopes:

$$S_K = \sum_{k=0}^{K} \left\lfloor \frac{x + 2^k}{2^{k+1}} \right\rfloor = \lfloor x \rfloor - \left\lfloor \frac{x}{2^{K+1}} \right\rfloor.$$

As $K \to \infty$, $x/2^{K+1} \to 0$, and the answer depends on which side of 0 it approaches from.

- If $x \geq 0$: once $2^{K+1} > x$ we have $0 \leq x/2^{K+1} < 1$, so the last floor is 0 and $S_K = \lfloor x \rfloor$ from then on.
- If $x < 0$: once $2^{K+1} \geq -x$ we have $-1 \leq x/2^{K+1} < 0$, so the last floor is -1 and $S_K = \lfloor x \rfloor + 1$ from then on.

In either case only finitely many terms are non-zero, so the infinite sum is

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+2}{2^2} \right\rfloor + \cdots = \begin{cases} \lfloor x \rfloor, & x \geq 0, \\ \lfloor x \rfloor + 1, & x < 0. \end{cases}$$

For a non-negative integer x the sum is simply x.

Checks

Try $x = 7$. The terms are $\lfloor 4 \rfloor = 4$, $\lfloor \frac{9}{4} \rfloor = 2$, $\lfloor \frac{11}{8} \rfloor = 1$, $\lfloor \frac{15}{16} \rfloor = 0$, and every later term is 0. They add up to $7 = \lfloor 7 \rfloor$. Try $x = -\frac{5}{2}$: the terms are $\lfloor -\frac{3}{4} \rfloor = -1$, $\lfloor -\frac{1}{8} \rfloor = -1$, $\lfloor \frac{3}{16} \rfloor = 0$, and so on. The sum is $-2 = \lfloor -\frac{5}{2} \rfloor + 1$, which confirms that the negative case really does differ. For (ii), $n = 3$ and $x = 1.5$ give $1 + 1 + 2 = 4 = \lfloor 4.5 \rfloor$. A computer check on twenty thousand random rationals x and values $1 \leq n \leq 30$ confirmed every identity in the question, as well as the telescoped partial sums.

Common slips

- In (ii)(c), using (a) only to step upwards, which covers $x \geq 0$ but says nothing about negative x.
- In (iii)(b), assuming that $\lfloor x/2^{K+1} \rfloor \to 0$ for all x. For negative x it settles at -1, not 0.
- Writing $\lfloor a + b \rfloor = \lfloor a \rfloor + \lfloor b \rfloor$ when neither a nor b is an integer. Part (i) is precisely the case in which this is allowed.

Beyond STEP

The identity in (ii)(c) is Hermite's identity. It belongs to a family of floor-function results that are central to elementary number theory, for example Legendre's formula $\sum_{k \geq 1} \lfloor N/p^k \rfloor$ for the power of a prime p dividing $N!$. The sum in (iii)(b) counts something too: for a positive integer x, the kth term is the number of integers from 1 to x that are divisible by 2^k but not by 2^{k+1}, and every integer is counted exactly once. Davenport's *The Higher Arithmetic* is a good place to see these ideas at work.

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